Data Note | Synthetic Aircraft Availability Scenarios

Origin and coverage

1,500 entirely simulated observations, 14 columns. These are illustrative aircraft-availability scenarios, NOT actual Air Astana/FlyArystan flight logs, maintenance events or spare-engine inventories.

Published evidence

S1 official report: 22 unscheduled engine removals and up to 13 aircraft grounded in 2025 peak season. S4 reports a USD 42.3m estimated adverse EBITDAR impact, five additional A320-family leases, 13 leased spare engines, dynamic capacity management, and 208 engine swaps since January 2024. S2 offers global industry context; S3 summarizes company results.

Data cleaning and structure

data_raw.csv is intentionally imperfect simulated input; data_clean.csv is the prepared dataset. The Python notebook documents treatment of missing values, impossible negative values and inconsistent season labels. These defects were intentionally inserted, not discovered in airline data.

Tests

H0-1: peak vs off-peak mean modeled downtime; H0-2: with vs without spare engine mean modeled downtime. Welch two-sided t-tests. Results and p-values describe only the constructed simulation.

Interpretation limits

No company-level causal conclusions, real-world p-values, or actual savings estimates can be inferred from synthetic observations. The 2025 financial figures are descriptive and cannot calibrate row-level simulated downtime. S1, S3 and S4 are not independent underlying datasets.

Sources and access

S1 https://ir.airastana.com/media/8deacf8f13c3255/air-astana-ar-2025-eng-int.pdf
S2 https://simpleflying.com/pratt-whitney-gtf-engine-quietly-grounded-more-jets-crisis-decade/
S3 https://www.enginecowl.com/air-astana-2025/
S4 https://simpleflying.com/a-42-million-engine-nightmare-how-one-company-is-outsmarting-a-disaster-that-should-have-grounded-them/
Full text of S2-S4 provided in chat by user; access recorded 2026-11-09.